

Technical Specifications:

111915-01-PL 4-channel Terminal Strip Interface Board, with output of +/-12V differential power

The Earthquake Science Center's National Strong Motion Network (NSMN) now has 70 digitizer-recorders being readied for deployment to upgrade the recording systems in currently non-functioning building and bridge structural arrays. The "terminal boards" being ordered, as described in detail below, are a 100% required interface between the digitizer-recorders and the strong motion seismic sensors mounted on structural elements on the structures. The terminal board hardware, which is essentially an adaptor having no computing or processing capacity on its own, is proprietary and allows sensor wires routed to a central location within a structure to be connected directly to the digitizer-recorders. This purchase is to meet this critical need.

Functionality:

a. This board plugs in directly to the mil-spec sensor connectors on the front end of a Kinemetrics Multi-Channel Digitizer (Models Granite, Basalt and Obsidian). The board performs 2 primary tasks:

1. Routes sensor connector lines to a terminal strip where bare wires running from the seismic sensors (located throughout a building) can be electrically attached to the digitizer recording and power circuits. The terminal board has no processing capability whatsoever. It is a 100% analog device.
2. The terminal board supplies +/- 12-volt differential power to the seismic sensors which is a requirement for the older sensors we have installed in our building and bridge arrays to function properly.

Requirements:

A single terminal board must plug into the socket sensor connectors on the front end of a Kinemetrics Multi-Channel Digitizer (Models Granite, Basalt and Obsidian). Each terminal board plugs into 2 connectors: A 26-socket 3-seismic-channel

connector and a 15-socket 1-seismic-channel connector. The on-center distance between the 2 connectors is approximately 2 7/8 inches. The terminal board must convert +12V (sourced from the Granite or Basalt or Obsidian digitizer) to +/- 12 Volt

differential power required to power the older seismic sensors. The Digitizer Models Granite, Basalt and Obsidian that are currently on our shelves waiting to be deployed are not usable in our existing structures without this board.